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Subject :PYTHON

Batch :5

Mini Project: Dice Rolling Game using Python

1. Project Title

Dice Rolling Game using Python

2. Problem Statement

To design and implement a simple dice rolling game where a player competes against the computer over multiple rounds and the final winner is determined based on scores.

3. Objective

- To develop a simple game using Python
 - To understand loops and conditional statements
 - To use functions and recursion
 - To apply operators and user input handling
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4. Tools Used

- Python 3
 - VS Code / IDLE
-

5. Description

This project simulates a dice rolling game between a player and the computer. In each round, both roll a dice, and the one with the higher value wins the round. Scores are maintained throughout the game, and the final winner is declared. The game also allows replay using recursion.

6. Algorithm

- Start the program
 - Ask user for number of rounds
 - For each round:
 - Roll dice for player and computer
 - Compare results
 - Update scores
 - Display final scores
 - Declare winner
 - Ask user for replay
 - Repeat using recursion
-

7. Program (Python Code)

```
import random
```

```
def roll_dice():
```

```
    return random.randint(1, 6)
```

```
def show_result(player, computer):
```

```
    print(f"\nYou rolled: {player}")
```

```
    print(f"Computer rolled: {computer}")
```

```
    if player > computer:
```

```
        print("You win this round!")
        return "player"
    elif player < computer:
        print("Computer wins this round!")
        return "computer"
    else:
        print("It's a tie!")
        return "tie"

def play_game():
    player_score = 0
    computer_score = 0

    rounds = int(input("Enter number of rounds: "))

    for i in range(1, rounds + 1):
        print(f"\n--- Round {i} ---")

        input("Press Enter to roll the dice...")

        player_roll = roll_dice()
        computer_roll = roll_dice()

        winner = show_result(player_roll, computer_roll)

        if winner == "player":
            player_score += 1
        elif winner == "computer":
```

```
computer_score += 1
```

```
print("\n==== FINAL SCORE =====")
```

```
print(f"You: {player_score}")
```

```
print(f"Computer: {computer_score}")
```

```
if player_score > computer_score:
```

```
    print("You won the game!")
```

```
elif player_score < computer_score:
```

```
    print("Computer won the game!")
```

```
else:
```

```
    print("It's a draw!")
```

```
def replay():
```

```
    choice = input("\nPlay again? (yes/no): ").lower()
```

```
if choice == "yes":
```

```
    play_game()
```

```
    replay()
```

```
else:
```

```
    print("Game Over.")
```

```
play_game()
```

```
replay()
```

8. Sample Output

Enter number of rounds: 3

--- Round 1 ---

Press Enter to roll the dice...

You rolled: 4

Computer rolled: 2

You win this round!

--- Round 2 ---

Press Enter to roll the dice...

You rolled: 1

Computer rolled: 2

Computer wins this round!

--- Round 3 ---

Press Enter to roll the dice...

You rolled: 3

Computer rolled: 1

You win this round!

===== FINAL SCORE =====

You: 2

Computer: 1

You won the game!

Play again? (yes/no): no

Game Over.

9. Result

The dice rolling game successfully simulates multiple rounds and determines the final winner based on scores.

10. Conclusion

This project demonstrates how Python can be used to build a simple interactive game using functions, loops, operators, and recursion.